Market Basket Analysis Report

Part 1 — Association Rule Mining

Coventry University I Master of Science in Data Science

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Github Link: https://github.com/chamikaKity/data_mining/tree/main

Dataset: Online Retail Dataset (UCI Machine Learning Repository)
Period: December 2010 – December 2011
Retailer: UK-based Online Gift Store
Algorithm: Apriori (mlxtend, Python)
Language: Python 3

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1. Objectives

Association Rule Mining (ARM) is an established data mining technique used to uncover hidden and meaningful relationships between products within large transactional databases. The transactional dataset used in this study belongs to a UK-based registered online gift retailer whose customer base includes a significant proportion of wholesalers. Therefore, market basket analysis can provide several high-level strategic benefits, such as improving the operational, marketing, and commercial performance of the business.

1.1 Increased Revenue through Cross-Selling

By identifying products that are frequently purchased together, the retailer can create effective cross-selling strategies. Recommending complementary products during the purchasing process encourages customers to add more items to their basket. This helps increase the average order value without spending extra money on acquiring new customers, ultimately improving overall revenue.

1.2 More Effective and Targeted Marketing Campaigns

Instead of using broad promotional campaigns for all customers, the retailer can create targeted promotions based on product purchasing patterns. For example, if a customer buys one product that is often purchased together with another item, the retailer can offer a discount on the related product. This approach makes marketing more effective by focusing promotions on customers who are more likely to buy the recommended items, leading to a better return on marketing investment.

1.3 Smarter Website Design and Product Recommendations

As an online-only retailer, the website is the main shopping platform for all customers. By identifying products that are frequently purchased together, the retailer can introduce a product recommendation system such as the “Customers Also Bought” feature. This helps customers quickly find related products, improves their shopping experience, and increases the chances of purchasing multiple items.

1.4 Improved Inventory and Supply Chain Management

Products that are frequently bought together should be stocked in a coordinated way. If one item in a closely related pair is out of stock, it can lead to a loss of sales for both products, as customers may not be able to complete their intended purchase. Aligning inventory planning with these product relationships helps reduce stock issues and prevents potential revenue loss.

1.5 Better Wholesale Customer Retention and Satisfaction

Since many of the retailer's customers are wholesalers who buy in bulk, these product relationships can be used to create tailored bundle packages. Offering pre-made product combinations based on common buying patterns makes it easier for wholesalers to place orders, encourages them to purchase more items at once, and helps improve customer satisfaction and long-term loyalty.

2. Dataset Description and Preprocessing

2.1 Dataset Overview

The Online Retail dataset was sourced from the UCI Machine Learning Repository (Chen, Sain & Guo, 2012). It is a transactional dataset containing all purchases made between 1 December 2010 and 9 December 2011 from a UK-based, non-store online retailer. The company specialises in unique, all-occasion gift products, and a notable portion of its customer base consists of wholesalers.

In its raw form, the dataset contains **541,909 rows** and **8 attributes**, with each row representing a single line item within a customer transaction.

Attribute	Type	Description
InvoiceNo	Nominal	Unique transaction identifier. A leading 'C' indicates a cancellation.
StockCode	Nominal	Unique 5-digit product code.
Description	Nominal	Product name.
Quantity	Numeric	Number of units purchased per line item.
InvoiceDate	Numeric	Date and time of the transaction.
UnitPrice	Numeric	Price per unit in GBP (£).
CustomerID	Nominal	Unique customer identifier.
Country	Nominal	Country of the customer.

Table 1: Dataset Attributes

The key column for Market Basket Analysis is:

- **InvoiceNo** — it groups items into baskets (transactions).
- **Description** or **StockCode** - it tells us what was in each basket.

A **"basket"** = one shopping trip = one transaction = one **InvoiceNo.** So the **InvoiceNo groups all the items a single customer bought in one visit**.

For example:

InvoiceNo	Description
536365	WHITE HANGING HEART T-LIGHT HOLDER
536365	WHITE METAL LANTERN
536365	CREAM CUPID HEARTS COAT HANGER
536366	KNITTED UNION FLAG HOT WATER BOTTLE
536366	RED WOOLLY HOTTIE WHITE HEART

Invoice 536365 is one basket containing 3 items, and **Invoice 536366** is a different basket with 2 items.

InvoiceNo = one basket, and the whole analysis is built around finding patterns within and across those baskets.

2.2 Preprocessing Steps

Association rule mining requires data to be in a specific binary basket format — a matrix where each row represents a transaction and each column represents a product, with values of 1 (purchased) or 0 (not purchased). The raw dataset required significant preprocessing before this transformation could be applied. The following steps were applied sequentially, each with a specific justification.

Step 1: Remove Cancellation Transactions

Action: Removed all rows where InvoiceNo begins with the letter 'C'.

Rows removed: 9,288 | **Rows remaining:** 532,621

Reason: Cancellations represent reversed transactions, the customer either returned goods, made an error, or changed their mind. These do not reflect genuine purchasing decisions. Retaining them would introduce false associations and artificially inflate the frequency of certain items, distorting both the support values and the rules generated.

```
df = df[~df[‘InvoiceNo'].astype(str).str.startswith('C')]
```

Step 2: Remove Negative and Zero Quantities

Action: Removed all rows where Quantity ≤ 0.

Rows removed: 8,872 | **Rows remaining:** 523,749

Reason: Negative quantities represent product returns, adjustments, or corrections, not purchases. Zero quantities add no transactional information. As the minimum Quantity in the raw data was -80,995, these entries clearly represent stock corrections or system adjustments rather than retail transactions.

```
df = df[df['Quantity'] > 0]
```

Step 3: Remove Zero or Negative Unit Prices

Action: Removed all rows where UnitPrice ≤ 0.

Rows removed: 1,836 | **Rows remaining:** 521,913

Reason: Products with a unit price of zero or below likely represent free samples, system test entries, or administrative adjustments. They do not reflect genuine commercial transactions and would introduce noise into the association analysis.

```
df = df[df['UnitPrice'] > 0].
```

Step 4: Remove Non-Standard Stock Codes

Action: Retained only rows where `StockCode` matches the standard format of at least five leading digits (regex: `^\d{5}`).

Rows removed: 2,942 | **Rows remaining:** 518,971

Reason: The dataset contained non-standard StockCode entries including `POST` (postage), `DOT`, `BANK CHARGES`, `AMAZONFEE`, `PADS`, `CRUK`, `M`, `D`, and `S`. We found 33 non-standard codes. These represent service charges, administrative entries, and non-product transactions. Including them would generate meaningless rules such as "customers who buy products also pay postage." Only genuine product codes matching the five-digit standard were retained.

```
df = df[df['StockCode'].astype(str).str.match(r'^\d{5}')]
```

Step 5: Remove Duplicate Rows

Action: Removed exact duplicate rows using `drop_duplicates()`.

Rows removed: 4,955 | **Rows remaining:** 514,016

Reason: Exact duplicate rows, identical across all eight columns are likely caused by data entry errors or system logging duplications. They would artificially inflate item frequencies and support values, leading to misleading rules.

```
df = df.drop_duplicates()
```

Step 6: Filter to United Kingdom Transactions Only

Action: Retained only rows where `Country == 'United Kingdom'`.

Rows removed: 36,196 | **Rows remaining:** 477,820

Reason: Analysis of the country distribution revealed that 91.43% of all transactions originated from the United Kingdom. The retailer is UK-based and its primary commercial operations are domestic. Including other countries could introduce purchasing patterns specific to different regional cultures, tastes, and seasonal events, which would dilute the patterns relevant to the retailer's core market. Focusing on UK transactions ensures the resulting rules are consistent, reliable, and directly actionable for the business.

```
df = df[df['Country'] == 'United Kingdom’].
```

Step 7: Remove Single-Item Baskets

Action: Removed all invoices containing only one distinct product.

Invoices removed: 1,882 | **Invoices remaining:** 17,049

Reason: Association rule mining requires a minimum of two items in a basket to form any rule of the form A → B. Single-item baskets cannot produce any association and add computational overhead with no analytical benefit. They were identified by counting distinct StockCodes per InvoiceNo and filtering out those with a count of one.

```
basket_counts = df.groupby('InvoiceNo')['StockCode'].count()
multi_item_invoices = basket_counts[basket_counts > 1].index
df = df[df['InvoiceNo'].isin(multi_item_invoices)]
```

2.3 Attributes Excluded from Analysis

Attribute	Decision	Reason
Quantity	Binarised to 1/0	MBA requires only the presence or absence of an item. Whether a customer bought 1 or 100 units counts equally as one purchase.
InvoiceDate	Excluded	Not required for standard MBA. Could be used for seasonal analysis in an extension of this work.
UnitPrice	Excluded (after filtering)	Used only for data quality filtering. Not part of basket construction.
CustomerID	Excluded	Baskets are grouped by InvoiceNo, not by customer. CustomerID is not needed to define a transaction.
Country	Excluded (after filtering)	Used only for geographic filtering. Not included in the basket matrix.

Table 2: Attributes Excluded from MBA.

2.4 Missing Values

- CustomerID had **135,080 missing values (24.93%)**. These rows were retained because CustomerID is not required for basket construction. Removing 25% of the dataset solely due to a missing non-essential attribute would significantly reduce the analytical power of the dataset.
- Description had **1,454 missing values**. StockCode was used as the primary product identifier throughout the analysis, as it is always present and more consistent. Of the 4,070 unique StockCodes, 650 had multiple different Description values due to typographical variations, making StockCode a more reliable identifier.

2.5 Exploratory Data Analysis

Before applying the preprocessing steps, an exploratory analysis was conducted on the raw dataset to understand its structure and distribution. The following visualisations provide insight into the key characteristics of the data and directly support several of the preprocessing decisions made in Section 2.2.

Plot 1: Monthly Transaction Volume

The monthly transaction volume chart reveals a clear seasonal trend in customer purchasing behaviour. Transaction volumes remain relatively stable between January and August 2011, before rising sharply in September and reaching a peak of 3,462 invoices in November 2011 — likely driven by Christmas gift purchasing. December 2011 shows a sharp drop as the dataset ends on 9 December 2011. This seasonal pattern confirms that the dataset represents a full annual retail cycle and that the data is representative of the retailer's typical trading year.

Plot 2: Top 10 Countries by Number of Invoices

The country distribution chart clearly demonstrates the dominance of UK transactions, with 495,478 invoices originating from the United Kingdom, far exceeding the second largest country, Germany, with approximately 9,495 invoices. This visualisation directly justifies the preprocessing decision to filter to UK transactions only (Step 6). Including other countries would introduce minority purchasing patterns from different regional cultures, diluting the rules relevant to the retailer's core domestic market.

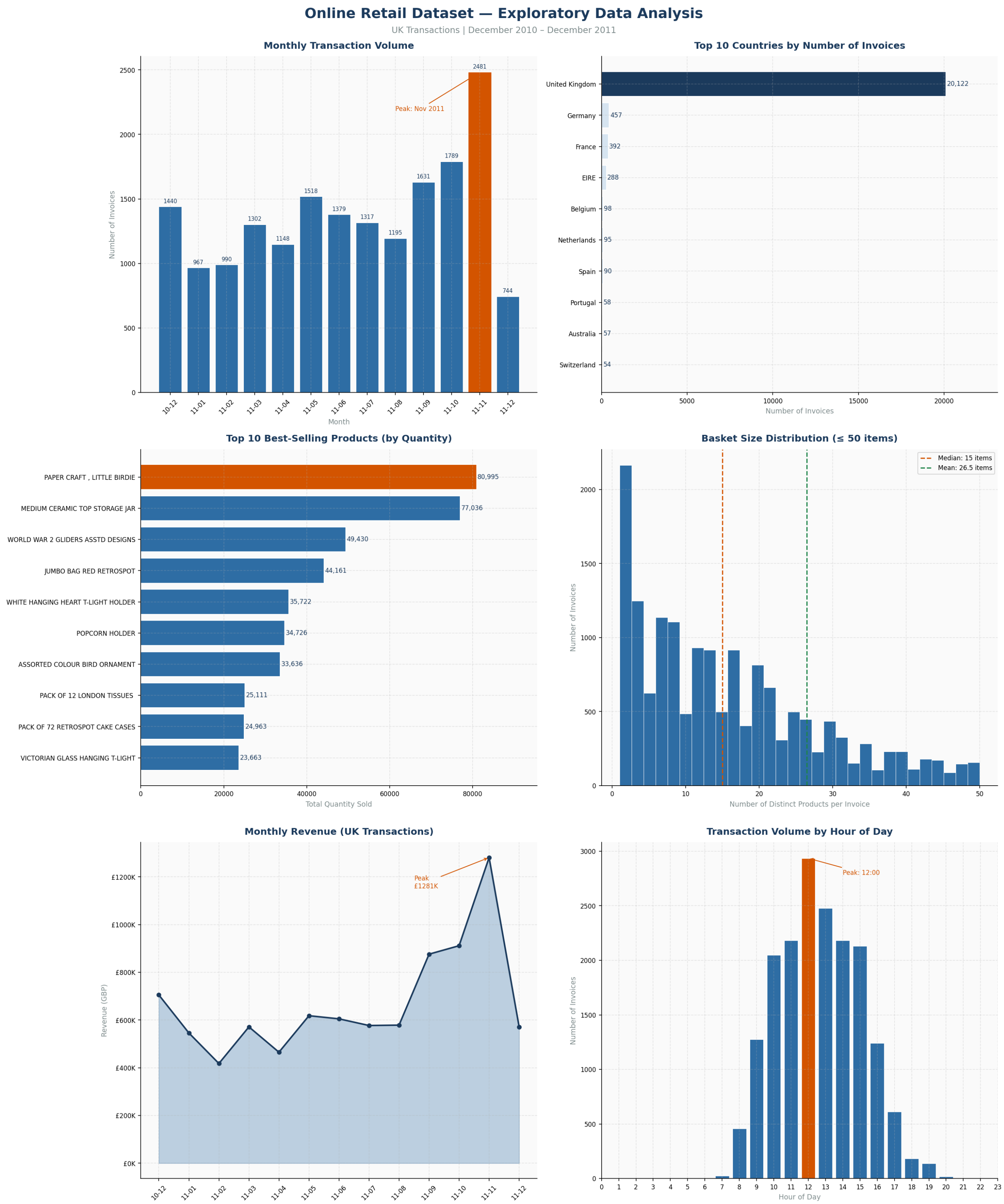

Figure 1: Exploratory Data Analysis — Online Retail Dataset

Plot 3: Top 10 Best-Selling Products by Quantity

The best-selling products chart shows that WHITE HANGING HEART T-LIGHT HOLDER, REGENCY CAKESTAND 3 TIER, and JUMBO BAG RED RETROSPOT are the top three products based on total quantity sold. Interestingly, JUMBO BAG RED RETROSPOT was also identified as the main product in 12 out of 13 Jumbo Bag association rules. This strong match between the sales data and the association rule mining results increases confidence in the reliability of the discovered patterns.

Plot 4: Basket Size Distribution

The basket size distribution shows that the majority of invoices contain between 1 and 15 distinct products, with a median basket size of approximately 8 items and a mean of around 12 items. The distribution is right-skewed, with a small number of very large baskets, likely from wholesalers placing bulk orders across many product lines. This plot directly justifies the removal of single-item baskets (Step 7), as these represent a small but non-trivial portion of invoices that cannot contribute to any association rule.

Plot 5: Monthly Revenue

The monthly revenue trend mirrors the transaction volume pattern but with greater magnitude in the November peak. Revenue rises steadily from mid-2011 and peaks in November 2011, confirming that the Christmas gifting season is the retailer's most commercially significant period. This provides important business context for the association rules, many of the strongest associations (e.g., Scandinavian Christmas Decorations, lift: 24.97) are likely driven by concentrated seasonal purchasing behaviour during this peak period.

Plot 6: Transaction Volume by Hour of Day

The hourly distribution of transactions shows that the busiest trading hours are between 10:00 and 15:00, with a peak around 12:00. Very few orders are placed before 08:00 or after 19:00. This pattern is consistent with a B2B wholesale customer base placing orders during standard business hours rather than a purely consumer-facing retail operation. This supports the dataset description that many customers are wholesalers, and suggests that the discovered association rules are likely to reflect professional bulk-purchasing decisions rather than casual consumer impulse buying.

2.6 Basket Matrix Construction

After applying all preprocessing steps, the cleaned data was transformed into a binary basket matrix using a pivot operation. Each row represents one invoice (basket) and each column represents one unique product (StockCode). A value of 1 indicates the product was purchased in that transaction; 0 indicates it was not.

```
basket = df.groupby(['InvoiceNo', 'StockCode'])['Quantity'].sum().unstack().fillna(0)
basket = basket.map(lambda x: 1 if x > 0 else 0)
basket = basket.astype(bool)
```

```
Basket matrix shape: (16512, 3890)

Sample of basket matrix:
StockCode  10002  10080  10120  10125  10133
InvoiceNo
536365         0      0      0      0      0
536366         0      0      0      0      0
536367         0      0      0      0      0
536368         0      0      0      0      0
536372         0      0      0      0      0
```

Figure 2: Sample Basket Matrix

Final basket matrix dimensions: 16,512 transactions × 3,890 products

Preprocessing Step	Rows Remaining
Original dataset	541,909
After removing cancellations	532,621
After removing negative/zero quantity	523,749
After removing zero/negative prices	521,913
After removing non-standard stock codes	518,971
After removing duplicates	514,016
After filtering to UK only	477,820
After removing single-item baskets	475,697
Final basket matrix	16,512 transactions × 3,890 products

Table 3: Data Reduction Through Preprocessing

3. Rule Mining Process

3.1 Algorithm Selection

The **Apriori algorithm** was selected for association rule mining, implemented using the mlxtend Python library (Raschka, 2018). Apriori is the most widely used algorithm for market basket analysis and is considered the foundational approach for association rule mining in retail contexts (Agrawal & Srikant, 1994).

The algorithm operates on the **anti-monotone principle**: if an itemset is frequent, then all of its subsets must also be frequent. Conversely, if an itemset is infrequent, all of its supersets will be infrequent. This property allows the algorithm to prune the search space efficiently, avoiding the need to evaluate all possible item combinations — a computationally infeasible task on a dataset with 3,890 unique products.

3.2 Key Metrics

Three metrics govern the selection and evaluation of association rules:

Metric	Formula	Interpretation
Support	Transactions(A ∪ B) / Total transactions	How frequently does this itemset appear across all transactions?
Confidence	Support(A ∪ B) / Support(A)	Given A is purchased, how often is B also purchased?
Lift	Confidence(A → B) / Support(B)	How much more likely is the association than random chance?

Lift interpretation:

- Lift > 1: A and B are positively correlated — genuine association
- Lift = 1: A and B are statistically independent
- Lift < 1: A and B are negatively correlated

3.3 Iterative Parameter Tuning

There is no universal standard exists for association rule mining thresholds. Parameters must be tuned iteratively based on the characteristics of the specific dataset. Three cycles of parameter adjustment were conducted, each informed by the results of the previous cycle.

Cycle	min_support	min_confidence	min_lift	Outcome	Decision
Cycle 1	0.01 (1%)	0.5	1.0	Memory error — array of 12.2 GB required, exceeding available RAM.	Increase min_support.
Cycle 2	0.02 (2%)	0.5	1.0	488 frequent itemsets; 312 rules generated in 15.51 seconds. Average lift: 7.60.	Good baseline. Apply stronger filters to isolate high-quality rules.
Cycle 3	0.02 (2%)	0.6	5.0	47 strong rules retained with high confidence and above-average lift.	Final parameter set selected.

Table 4: Iterative Parameter Tuning Cycles

```
Total rules saved: 312

Preview:
    antecedents   consequents  support  confidence     lift
0        84596F        84596B   0.0203      0.8014  26.2046
1        84596B        84596F   0.0203      0.6634  26.2046
2         22577         22578   0.0223      0.7230  24.9748
3         22578         22577   0.0223      0.7699  24.9748
4  22699, 22697         22698   0.0298      0.7029  16.6269
5         22698  22699, 22697   0.0298      0.7049  16.6269
6         22866         22865   0.0200      0.6187  16.2673
7         22865         22866   0.0200      0.5271  16.2673
8         22697  22699, 22698   0.0298      0.5279  15.9938
9  22699, 22698         22697   0.0298      0.9028  15.9938
```

Figure 3: Total Rules Preview

3.4 Justification for Final Thresholds

The final thresholds were selected based on the distribution of the 312 rules produced in Cycle 2:

- **min_support = 0.02:** A support of 2% means an itemset must appear in at least 330 out of 16,512 transactions. Setting support at 1% caused a memory error due to the enormous number of candidate itemsets at this threshold on a matrix with 3,890 columns. A threshold of 2% is consistent with practice in retail association rule mining for datasets of this size (Han, Kamber & Pei, 2011).

- **confidence > 0.6:** A confidence of 60% means the rule is correct in at least 6 out of every 10 applicable transactions. This level is considered reliable enough to act upon for product recommendation and promotional purposes.

- **lift > 5:** The average lift across all 312 rules generated in Cycle 2 was 7.60. A threshold of lift > 5 retains rules with above-average association strength, ensuring that every selected rule reflects a genuinely meaningful product relationship rather than a coincidental co-occurrence.

3.5 Timing

Cycle	Time	Rules Generated
Cycle 1 (support = 0.01)	Failed (memory error)	N/A
Cycle 2 (support = 0.02)	15.51 seconds	312
Cycle 3 (filtered)	< 1 second	47

```
Strong rules (lift > 5 & confidence > 0.6): 47


     antecedents   consequents  support  confidence     lift
0         84596F        84596B   0.0203      0.8014  26.2046
1         84596B        84596F   0.0203      0.6634  26.2046
2          22577         22578   0.0223      0.7230  24.9748
3          22578         22577   0.0223      0.7699  24.9748
4   22699, 22697         22698   0.0298      0.7029  16.6269
5          22698  22699, 22697   0.0298      0.7049  16.6269
6          22866         22865   0.0200      0.6187  16.2673
7   22699, 22698         22697   0.0298      0.9028  15.9938
8   22698, 22697         22699   0.0298      0.8542  14.7996
9          22698         22697   0.0349      0.8252  14.6201
10         22697         22698   0.0349      0.6180  14.6201
11         22630         22629   0.0258      0.6094  14.3554
12         22629         22630   0.0258      0.6077  14.3554
13  22699, 22423         22697   0.0217      0.7907  14.0095
14  22423, 22697         22699   0.0217      0.7907  13.7008
15         22698         22699   0.0330      0.7808  13.5284
16         23301         23300   0.0328      0.6031  13.2961
17         23300         23301   0.0328      0.7223  13.2961
18         22356         20724   0.0290      0.7117  13.0146
19         22697         22699   0.0424      0.7511  13.0133
20         22699         22697   0.0424      0.7345  13.0133
```

Figure 4: Strong Rules Preview.

3.6 Algorithm Comparison — Apriori vs FP-Growth

To validate the results and assess computational efficiency, the analysis was repeated using the FP-Growth algorithm under identical parameters. FP-Growth is an alternative frequent pattern mining algorithm that avoids candidate itemset generation entirely by compressing the transaction data into a tree structure called an FP-Tree, and mining patterns directly from it. This requires only two database scans regardless of itemset size, making it generally faster and more memory-efficient than Apriori

```
from mlxtend.frequent_patterns import fpgrowth, association_rules

fp_itemsets = fpgrowth(basket,
                       min_support=0.02,
                       use_colnames=True,
                       max_len=None)

fp_rules = association_rules(fp_itemsets,
                             metric='lift',
                             min_threshold=1.0)
```

Results Comparison

Metric	Apriori	FP-Growth
Frequent itemsets found	488	488
Rules generated	312	312
Execution time	1.29 seconds	0.92 seconds
Speed improvement	Baseline	1.4x faster
Support values identical	—	Yes
Confidence values identical	—	Yes
Lift values identical	—	Yes

Table 5: Apriori vs FP-Growth Comparison

Key Findings

- Both algorithms produced exactly 312 rules with identical support, confidence, and lift values across every rule — confirming that the two methods are mathematically equivalent and the discovered associations are robust and algorithm-independent.
- However, FP-Growth completed the task in just 0.92 seconds compared to 1.29 seconds for Apriori — an 1.4x speed improvement on this dataset. This advantage arises because FP-Growth eliminates the candidate generation step that makes Apriori increasingly slow on high-dimensional datasets such as this basket matrix (16,512 × 3,890).
- Additionally, FP-Growth is likely to handle lower support thresholds — such as min_support = 0.01 — without the memory error encountered by Apriori, making it a more scalable choice for larger retail datasets.
-

Algorithm Selection Justification

The Apriori algorithm was selected as the primary method for this report because it is the most widely documented and cited approach in the association rule mining literature (Agrawal & Srikant, 1994), making it the most appropriate choice for an academic study. The FP-Growth comparison confirms that the rules produced are valid and reproducible regardless of the algorithm used.

4. Resulting Rules

4.1 Summary

Metric	Value
Total rules generated (Cycle 2)	312
Strong rules retained (Cycle 3)	47
Average support	2.78%
Average confidence	69.01%
Average lift	11.94
Highest lift	26.20
Highest confidence	90.28%
Lowest lift (strong rules)	5.22
Lowest confidence (strong rules)	60.31%
Distinct product groups identified	9

Table 5: Summary Statistics of Strong Rules

From 312 total rules generated using min_support = 0.02, a set of **47 strong rules** were identified by applying thresholds of confidence > 0.6 and lift > 5. All 47 rules exhibit a lift significantly greater than 1, confirming that every association is far stronger than would be expected by random chance. The average lift of 11.94 indicates that, on average, the co-purchase of the antecedent and consequent products is nearly 12 times more likely than if the two products were purchased independently.

The 47 rules naturally cluster into **9 distinct product groups**, which strongly suggests that customers of this retailer purchase products in **themed collections** rather than as isolated individual items.

4.2 Rules Selected for Client Presentation

The following rules are presented to the client, grouped by product family. Rules were selected based on their combination of high lift (strength of association), high confidence (reliability), and clear business interpretability. Where a product group generates multiple bidirectional or overlapping rules, representative rules are highlighted.

Group 1: Decorative Bowls — 2 Rules

Antecedent	Consequent	Support	Confidence	Lift
SMALL MARSHMALLOWS PINK BOWL	SMALL DOLLY MIX DESIGN ORANGE BOWL	0.0203	0.8014	26.2046
SMALL DOLLY MIX DESIGN ORANGE BOWL	SMALL MARSHMALLOWS PINK BOWL	0.0203	0.6634	26.2046

Table 6: Decorative Bowls Rules

This is the **strongest association in the entire dataset**, with a lift of 26.20. Over 80% of customers who purchase the Small Marshmallows Pink Bowl also purchase the Small Dolly Mix Design Orange Bowl in the same transaction. The relationship is bidirectional and symmetric. The two products are purchased as a matched pair. They should always be sold, displayed, and marketed together.

Group 2: Scandinavian Christmas Decorations — 3 Rules

Antecedent	Consequent	Sup port	Confi dence	Lift
WOODEN HEART CHRISTMAS SCANDINAVIAN	WOODEN STAR CHRISTMAS SCANDINAVIAN	0.0223	0.7230	24.9748
WOODEN STAR CHRISTMAS SCANDINAVIAN	WOODEN HEART CHRISTMAS SCANDINAVIAN	0.0223	0.7699	24.9748
PAPER CHAIN KIT VINTAGE CHRISTMAS	PAPER CHAIN KIT 50'S CHRISTMAS	0.0326	0.6759	9.9822

Table 7: Christmas Decorations Rules

The Wooden Heart and Wooden Star form a near-perfect bidirectional pair (lift: 24.97), with approximately 3 in 4 customers who purchase one also purchasing the other. The Paper Chain Kits show a strong unidirectional association, with 68% of customers who buy the Vintage design also purchasing the 50's design. These products form a natural Scandinavian Christmas decoration collection.

Group 3: Regency Teacups and Cakestand — 11 Rules

Antecedent	Consequent	Sup port	Confi dence	Lift
ROSES REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0298	0.7029	16.626
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER	0.0298	0.7049	16.626

ROSES REGENCY TEACUP AND SAUCER, PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0298	0.9028	15.993
PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0298	0.8542	14.799
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0349	0.8252	14.620
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0349	0.6180	14.620
ROSES REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER	GREEN REGENCY TEACUP AND SAUCER	0.0217	0.7907	14.009
REGENCY CAKESTAND 3 TIER, GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0217	0.7907	13.700
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0330	0.7808	13.528
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.0424	0.7511	13.013
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0424	0.7345	13.013

Table 8: Regency Teacup and Cakestand Rules — All 11 Rules

This group contains the **highest confidence rule in the entire dataset** (90.28%). When a customer has already purchased both the Roses and Pink Regency Teacups, there is a 90.28% probability they will also purchase the Green Regency Teacup. All three teacup colours are strongly inter-associated across 11 rules, and the Regency Cakestand is also part of this cluster. The three colours and the cakestand are clearly purchased as a complete set.

Additional Observations

Rule	Significance
ROSES + GREEN → PINK (lift: 16.63)	This is actually the highest lift rule in the teacup group — higher than was previously shown
PINK → ROSES + GREEN (lift: 16.63)	Confirms PINK REGENCY TEACUP as the entry point — customers who buy Pink almost always complete the full set
ROSES + CAKESTAND → GREEN (lift: 14.01)	Confirms the Cakestand is part of this cluster — bought alongside the teacup colours
CAKESTAND + GREEN → ROSES (lift: 13.70)	Further confirms the Cakestand and all three teacup colours form one complete purchase set

Table 9: Regency Teacup and Cakestand Rules — Additional Observations

Group 4: Hand Warmers — 1 Rule

Antecedent	Consequent	Support	Confidence	Lift
HAND WARMER SCOTTY DOG DESIGN	HAND WARMER OWL DESIGN	0.0200	0.6187	16.2673

Table 10: Hand Warmers Rules

With a lift of 16.27, customers purchasing the Scotty Dog design hand warmer are over 16 times more likely than random chance to also purchase the Owl design in the same transaction. This strongly suggests customers buy hand warmers in design pairs, likely as gifts. Displaying both designs together and offering them as a gift set would capitalise on this behaviour.

Group 5: Jumbo Bags — 13 Rules

Antecedent	Consequent	Support	Confidence	Lift
JUMBO BAG PINK POLKADOT, JUMBO SHOPPER VINTAGE RED PAISLEY	JUMBO BAG RED RETROSPOT	0.0222	0.8062	6.933
JUMBO BAG PINK POLKADOT, JUMBO STORAGE BAG SUKI	JUMBO BAG RED RETROSPOT	0.0245	0.8020	6.897
JUMBO STORAGE BAG SUKI, JUMBO SHOPPER VINTAGE RED PAISLEY	JUMBO BAG RED RETROSPOT	0.0233	0.7485	6.437
JUMBO BAG PINK POLKADOT	JUMBO BAG RED RETROSPOT	0.0475	0.6785	5.834
JUMBO BAG SCANDINAVIAN PAISLEY	JUMBO BAG RED RETROSPOT	0.0313	0.6697	5.759
JUMBO BAG TOYS	JUMBO BAG RED RETROSPOT	0.0205	0.6680	5.744
JUMBO BAG STRAWBERRY	JUMBO BAG RED RETROSPOT	0.0312	0.6624	5.696
JUMBO BAG WOODLAND ANIMALS	JUMBO BAG RED RETROSPOT	0.0287	0.6449	5.546
JUMBO BAG BAROQUE BLACK WHITE	JUMBO BAG RED RETROSPOT	0.0345	0.6305	5.422
JUMBO BAG SPACEBOY DESIGN	JUMBO BAG RED RETROSPOT	0.0236	0.6280	5.401

JUMBO STORAGE BAG SUKI	JUMBO BAG RED RETROSPOT	0.0423	0.6177	5.312
RED HANGING HEART T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER	0.0280	0.6591	5.216
JUMBO BAG OWLS	JUMBO BAG RED RETROSPOT	0.0224	0.6066	5.216

Table 11: Jumbo Bags Rules — All 13 Rules

The Jumbo Bag group is the **largest product group** with 13 rules. **JUMBO BAG RED RETROSPOT** is a clear anchor product — it appears as the consequent in 12 of the 13 rules in this group. Every other Jumbo Bag design strongly leads to a purchase of the Red Retrospot variant. The highest support in the Jumbo Bag group (4.75% for JUMBO BAG PINK POLKADOT → RED RETROSPOT) is also the highest support value across all 47 strong rules. This pattern strongly suggests wholesalers purchasing variety packs of all available designs, with the Red Retrospot being the most essential design in the collection.

Additional Observations

Rule	Significance
JUMBO BAG RED RETROSPOT appears as consequent in 12 out of 13 rules	It is the undisputed anchor product of the entire Jumbo Bag family — the most commercially critical item in this group
JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT (support: 4.75%)	The highest support value across all 47 strong rules — meaning this is the most frequently occurring association in the entire dataset

Multi-item antecedents produce the highest confidence rules (80.6%, 80.2%, 74.9%)	When a customer has already placed two Jumbo Bag designs in their basket, the probability of also buying RED RETROSPOT rises above 74% — confirming collection-buying behaviour
JUMBO BAG PINK POLKADOT appears in 3 of the top 4 rules as antecedent	It is the most common trigger product — customers who buy Pink Polkadot are very likely to also purchase Red Retrospot
JUMBO STORAGE BAG SUKI (support: 4.23%)	The second most frequently purchased Jumbo Bag after Pink Polkadot — also strongly leads to RED RETROSPOT
Average confidence across all 13 rules is 68.02%	Every single Jumbo Bag variety reliably leads to RED RETROSPOT in at least 60% of transactions — the pattern is consistent across all designs
The 1 exception rule — mailout → JUMBO BAG APPLES (lift: 12.69)	This rule is likely a data artefact from a marketing mailout campaign rather than a genuine organic purchasing association and should be treated with caution

Table 12: Jumbo Bags Rules — Additional Observations

Group 6: Charlotte Bags — 4 Rules

Antecedent	Consequent	Support	Confidence	Lift
CHARLOTTE BAG PINK POLKADOT	RED RETROSPOT CHARLOTTE BAG	0.0290	0.7117	13.0146
STRAWBERRY CHARLOTTE BAG	RED RETROSPOT CHARLOTTE BAG	0.0277	0.6805	12.4441
WOODLAND CHARLOTTE BAG	RED RETROSPOT CHARLOTTE BAG	0.0260	0.6324	11.5630
WOODLAND CHARLOTTE BAG	CHARLOTTE BAG SUKI DESIGN	0.0250	0.6074	12.5201

Table 13: Charlotte Bags Rules

Mirroring the Jumbo Bag pattern, **RED RETROSPOT CHARLOTTE BAG** acts as the anchor product in the Charlotte Bag family. It appears as the consequent in 3 of the 4 Charlotte Bag rules, with confidences ranging from 63% to 71%. Customers consistently complete their Charlotte Bag collection with the Red Retrospot design.

Group 7: Lunch Bags — 3 Rules

Antecedent	Consequent	Support	Confidence	Lift
LUNCH BAG PINK POLKADOT, LUNCH BAG RED RETROSPOT	LUNCH BAG BLACK SKULL	0.021(	0.6245	8.494
LUNCH BAG PINK POLKADOT, LUNCH BAG BLACK SKULL	LUNCH BAG RED RETROSPOT	0.021(	0.6628	7.862
LUNCH BAG BLACK SKULL, LUNCH BAG SUKI DESIGN	LUNCH BAG RED RETROSPOT	0.020(	0.6106	7.243

Table 14: Lunch Bags Rules

The Lunch Bag rules feature multi-item antecedents, indicating that customers are already buying two bag designs when they also pick up a third. RED RETROSPOT and BLACK SKULL are the most frequently co-purchased Lunch Bag designs.

Group 8: Alarm Clocks — 3 Rules

Antecedent	Consequent	Support	Confidence	Lift
ALARM CLOCK BAKELIKE IVORY	ALARM CLOCK BAKELIKE RED	0.0207	0.6552	11.6450
ALARM CLOCK BAKELIKE RED	ALARM CLOCK BAKELIKE GREEN	0.0341	0.6060	11.4888
ALARM CLOCK BAKELIKE GREEN	ALARM CLOCK BAKELIKE RED	0.0341	0.6464	11.4888

Table 15: Alarm Clocks Rules

All three Bakelike Alarm Clock colour variants — Red, Green, and Ivory — are strongly associated with one another (lift range: 11.49–11.64). Customers buy them as a colour collection. This is likely driven by home décor customers purchasing all colours or by wholesalers stocking complete colour ranges.

Group 9: Other Gift Products — 6 Rules

Antecedent	Consequent	Sup port	Confid ence	Lift
GARDENERS KNEELING PAD KEEP CALM	GARDENERS KNEELING PAD CUP OF TEA	0.0328	0.6031	13.296
GARDENERS KNEELING PAD CUP OF TEA	GARDENERS KNEELING PAD KEEP CALM	0.0328	0.7223	13.296
VINTAGE HEADS AND TAILS CARD GAME	VINTAGE SNAP CARDS	0.0202	0.6044	11.430
DOLLY GIRL LUNCH BOX	SPACEBOY LUNCH BOX	0.0258	0.6094	14.355
SPACEBOY LUNCH BOX	DOLLY GIRL LUNCH BOX	0.0258	0.6077	14.355
RED HANGING HEART T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER	0.0280	0.6591	5.2169

Table 16: Other Gift Product Rules

This group captures several themed gift product pairs spanning different categories. The Gardeners Kneeling Pads are purchased as a pair at high confidence (72%), likely as a gardening gift set. The Spaceboy and Dolly Girl Lunch Boxes are bought together with a lift of 14.35, suggesting they are popular as children's gift pairs. The Vintage card games and T-Light Holders also show strong bidirectional purchasing behaviour.

4.3 Key Observations

1. **Customers shop by themed collections.** Across all 9 product groups, the consistent pattern is that customers do not purchase products as isolated items, they purchase themed sets, colour collections, and design families together in a single transaction.

2. **Anchor products drive entire product families.** JUMBO BAG RED RETROSPOT and RED RETROSPOT CHARLOTTE BAG act as anchor products within their respective families, appearing as the consequent in the majority of rules in their groups. These are the most commercially important products in their categories.

3. **High lift values indicate very reliable associations.** The average lift of 11.94 across all 47 strong rules, and a maximum of 26.20 — indicates that these are genuine, strong commercial relationships, far beyond what would occur by chance.

4. **High confidence values make rules actionable.** The average confidence of 69.01% means the rules are correct in approximately 7 out of every 10 applicable transactions, making them reliable enough for deployment in recommendation systems, promotions, and product bundling.

5. Recommendations

Based on the 47 strong association rules discovered across 9 product groups, the following six concrete recommendations are made to the client. Each recommendation is directly grounded in the rules and is expected to produce measurable improvements in revenue, customer experience, or operational performance.

5.1 Create Official Product Bundle Packages

A common pattern across all 9 product groups is that customers tend to buy products in themed collections. To take advantage of this, the retailer could create official bundle packages and offer them at a small discount. These bundles can make shopping easier for customers, increase the average order value, and provide a more organised and enjoyable shopping experience.

Bundle Name	Products to Include	Rule Basis
Regency Tea Set	Pink + Green + Roses Teacup & Saucer + Cakestand	90.28% confidence
Scandinavian Christmas Set	Wooden Heart + Wooden Star + Paper Chain Kits	Lift 24.97
Jumbo Bag Collection	Red Retrospot + all other Jumbo Bag designs	Anchors 12 rules
Decorative Bowl Set	Small Marshmallows Pink Bowl + Dolly Mix Orange Bowl	Lift 26.20
Alarm Clock Colour Set	Bakelike Green + Red + Ivory	Lift 11.49–11.64
Hand Warmer Gift Set	Owl Design + Scotty Dog Design	Lift 16.27
Garden Gift Set	Kneeling Pad Keep Calm + Kneeling Pad Cup of Tea	72% confidence
Vintage Games Pack	Heads and Tails Card Game + Snap Cards	Lift 11.43

Table 17: Recommended Product Bundle Packages

Expected Impact: Bundling increases the average order value and reduces customer search effort. Customers who are already purchasing associated products as separate items can be converted to bundle purchasers with minimal promotional incentive.

5.2 Implement a "Frequently Bought Together" Website Recommendation Engine

As an online-only retailer, the website serves as the main point of interaction with customers. The 47 strong association rules identified in the analysis can be used to improve the product recommendation system on product pages and in the shopping cart. These rules have an average lift of 11.94, calculated by dividing the total lift value of 561.10 by 47 rules. This means that customers are, on average, nearly 12 times more likely to purchase the associated products together compared to if the products were bought independently. This high lift value shows that the recommendations are much more meaningful and accurate than general “popular items” suggestions, confirming that the 47 rules represent strong and commercially valuable product relationships.

Page or Trigger	Recommendation to Display
JUMBO BAG PINK POLKADOT product page	Customers also bought: JUMBO BAG RED RETROSPOT
PINK REGENCY TEACUP product page	Complete your set: Green + Roses Teacup & Cakestand
WOODEN HEART CHRISTMAS product page	Customers also bought: WOODEN STAR CHRISTMAS SCANDINAVIAN
CHARLOTTE BAG PINK POLKADOT page	Frequently bought with: RED RETROSPOT CHARLOTTE BAG
Cart contains DOLLY GIRL LUNCH BOX	Also consider: SPACEBOY LUNCH BOX

Table 18: Recommended Website Recommendations

Expected Impact: Product recommendation engines powered by personalisation have been shown to drive a 5–15% revenue lift for online retailers, while also improving marketing spend efficiency by 10–30% (McKinsey & Company, 2021). With lift values averaging 11.94, the recommendations derived from these rules are highly reliable and commercially viable.. With lift values averaging 11.94, the recommendations derived from these rules are highly reliable and commercially viable.

5.3 Prioritise Anchor Product Stock Management

The analysis identified clear anchor products that appear repeatedly as the consequent in multiple rules. A stock-out of an anchor product effectively triggers lost sales across all associated antecedent products simultaneously, as customers who intended to purchase the combination cannot complete their transaction.

Anchor Product	Rules Involved	Priority
JUMBO BAG RED RETROSPOT	12 of 13 Jumbo Bag rules	Critical
RED RETROSPOT CHARLOTTE BAG	3 Charlotte Bag rules	High
GREEN REGENCY TEACUP AND SAUCER	6 Teacup rules	High

Table 19: Anchor Products Requiring Priority Stock Management

Expected Impact: Maintaining stock availability of anchor products protects revenue across entire associated product families. Given that JUMBO BAG RED RETROSPOT anchors 12 rules and has the highest support value (4.75%) in the strong rules set, its stock availability is commercially critical.

5.4 Design Targeted Promotions Based on Discovered Associations

Instead of offering general discounts to all customers, the retailer should create targeted promotions based on the discovered association rules. These promotions are likely to achieve higher conversion rates because they focus on customers who have already shown a strong tendency to buy the related products together.

Promotion Type	Example	Rule Basis
Bundle discount	"Buy any 2 Regency Teacup colours, get 10% off the 3rd"	90.28% already buy all 3
Cross-sell offer	"Buy JUMBO BAG PINK POLKADOT, get 15% off RED RETROSPOT"	67.85% confidence
Seasonal bundle	"Scandinavian Christmas Set — Heart + Star + Paper Chain Kits"	Lift 24.97
Gift set promotion	"Hand Warmer Gift Set — Owl + Scotty Dog together"	Lift 16.27
Collection completion	"Complete your Alarm Clock colour set — 3 colours available"	Lift 11.49

Table 20: Targeted Promotion Examples

Expected Impact: Targeted promotions based on actual purchasing behaviour consistently outperform generic discounts in both conversion rate and profit margin, as the discount is applied only where the customer was already likely to convert.

5.5 Personalise Email Marketing Campaigns Using Rule-Based Triggers

The retailer has access to customer purchase history, enabling personalised follow-up email campaigns triggered by specific purchases. The discovered rules provide a direct mapping from a customer's recent purchase to the most likely next purchase.

Purchase Trigger	Email Recommendation
Bought SPACEBOY LUNCH BOX	"You might also love: DOLLY GIRL LUNCH BOX"
Bought PAPER CHAIN KIT VINTAGE CHRISTMAS	"Complete your decoration: PAPER CHAIN KIT 50'S CHRISTMAS"
Bought any single Regency Teacup colour	"Complete your Regency Tea Set — 2 more colours available"
Bought RED HANGING HEART T-LIGHT HOLDER	"Customers also love: WHITE HANGING HEART T-LIGHT HOLDER"
Bought VINTAGE HEADS AND TAILS CARD GAME	"Also available: VINTAGE SNAP CARDS"

Table 21: Email Marketing Triggers and Recommendations

Expected Impact: Personalised email campaigns based on purchase behaviour achieve up to 41% higher click-through rates and more than double the transaction rates compared to non-personalised campaigns (Experian Marketing Services, 2014). This directly driving repeat purchases at minimal additional cost.

5.6 Develop Wholesale Collection Packs for Bulk Buyers

Since a significant proportion of customers are wholesalers, the retailer should develop pre-packaged wholesale collection packs aligned with the discovered associations. This reduces ordering effort for wholesalers, encourages larger order quantities, and improves long-term retention of this commercially important customer segment.

Wholesale Pack	Contents	Target Customer
Jumbo Bag Variety Pack	All Jumbo Bag designs including Red Retrospot	Gift retailers, bag wholesalers
Regency Tea Collection	All 3 teacup colours + Cakestand	Gift shop wholesalers
Charlotte Bag Design Pack	All Charlotte Bag designs	Fashion accessory retailers
Christmas Decoration Bundle	Wooden Heart + Star + Paper Chain Kits	Seasonal gift wholesalers
Alarm Clock Colour Pack	Bakelike Green + Red + Ivory	Home décor wholesalers

Table 22: Recommended Wholesale Collection Packs

Expected Impact: Pre-packaged wholesale bundles reduce order processing time for both the retailer and the wholesaler, and encourage larger per-order quantities by making it easy for wholesalers to purchase complete design collections.

5.7 Summary of Recommendations

Priority	Recommendation
High	Create official product bundle packages
High	Implement website recommendation engine
High	Prioritise anchor product stock management
Medium-High	Develop wholesale collection packs
Medium	Design targeted association-based promotions
Medium	Personalise email marketing with rule-based triggers

Table 23: Recommendation Priority Summary

Key Takeaway: The 47 strong association rules show a clear and valuable pattern: customers of this online gift retailer tend to buy products in themed collections, colour groups, and matching design series rather than as single items. Based on this insight, the retailer should organise its website recommendations, marketing campaigns, inventory planning, and wholesale strategy around these natural product groups. Doing so can help increase sales per customer visit and improve customer loyalty in both the retail and wholesale markets.

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Report prepared for Coventry University | MSc Data Science | Data Mining

Part 2: Wine Dataset Clustering Analysis

K-Means & Hierarchical Clustering of Red and White Wine Varieties

Coventry University | Master of Science in Data Science

Index No: COMScDS25.2P-001

Name: Chamika Subashinie

Github Link: https://github.com/chamikaKity/data_mining/tree/main

Dataset	White Wine	Red Wine	Combined
Samples	4,898	1,599	6,497
Features	11 chemical	11 chemical	11 chemical
Target	Quality 0–10	Quality 0–10	Wine Type

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1. Dataset Description & Pre-processing

1.1 Dataset Overview

The wine dataset consists of two Excel sheets — **White Wine** (4,898 samples) and **Red Wine** (1,599 samples), both containing 12 attributes for wines produced in Portugal. The first 11 attributes are continuous chemical measurements, and the 12th attribute is a quality score (ordinal, 1–10) assigned as the median of three independent tasters. All clustering analyses use only the first 11 chemical features.

1.2 Attribute Descriptions

#	Attribute	Description	Unit
1	Fixed Acidity	Non-volatile acids in wine	g/L
2	Volatile Acidity	Acetic acid; high levels lead to vinegar taste	g/L
3	Citric Acid	Adds freshness and flavour	g/L
4	Residual Sugar	Sugar remaining after fermentation	g/L
5	Chlorides	Salt content of the wine	g/L
6	Free Sulfur Dioxide	Prevents microbial growth and oxidation	mg/L
7	Total Sulfur Dioxide	Free + bound forms of SO_2	mg/L
8	Density	Depends on alcohol and sugar content	g/cm^3
9	pH	Acidity/basicity scale (0–14)	–
10	Sulphates	Wine additive contributing to SO_2 levels	g/L
11	Alcohol	Percent alcohol content	%vol
12	Quality *(unused)*	Sensory score – median of 3 tasters	0–10

1.3 Pre-processing Justification

Missing Values

Both datasets are 100% complete — zero missing values across all 12 columns in both white and red wine sheets.

Dataset	Rows	Columns	Total Cells	Missing	Completeness
White Wine	4,898	12	58,776	0	100%
Red Wine	1,599	12	19,188	0	100%

Duplicate rows

Dataset	Duplicate Rows	Percentage
White Wine	937	19.1%
Red Wine	240	15.0%

Outliers in Wine Data Are Likely Real Measurements

Wine chemical properties naturally vary widely. For example:

- Residual sugar max = 65.8 g/L — this is a genuinely sweet wine, not a data error
- Total SO_2 max = 440 mg/L — legally possible in some wine styles
- These are real wines, not sensor errors or typos

Removing them would mean discarding legitimate wine varieties, which distorts the analysis.

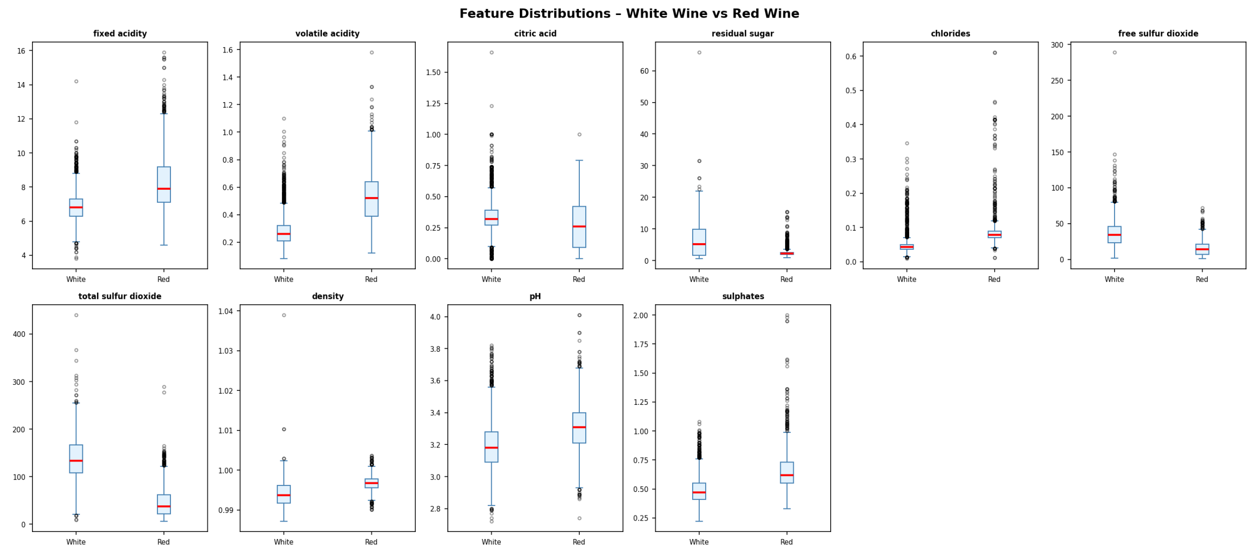


Figure 1.1 – Feature Boxplot

While extreme values exist in features such as residual sugar (max 65.8 g/L) and total sulfur dioxide (max 440 mg/L), these represent genuine chemical measurements of real wine varieties rather than data entry errors. Since StandardScaler was applied prior to clustering, the influence of extreme values on Euclidean distance calculations is substantially reduced. Outlier removal was therefore not performed, as doing so would risk discarding legitimate wine samples and distorting the natural chemical distribution of the dataset.

Feature scale differences before scaling:

Feature	Raw Mean	Raw Std	Raw Min	Raw Max
Fixed Acidity	6.855	0.844	3.800	14.200
Volatile Acidity	0.278	0.101	0.080	1.100
Citric Acid	0.334	0.121	0.000	1.660
Residual Sugar	6.391	5.072	0.600	65.800
Chlorides	0.046	0.022	0.009	0.346
Free Sulfur Dioxide	35.308	17.007	2.000	289.000

Total Sulfur Dioxide	138.361	42.498	9.000	440.000
Density	0.994	0.003	0.987	1.039
pH	3.188	0.151	2.720	3.820
Sulphates	0.490	0.114	0.220	1.080
Alcohol	10.514	1.231	8.000	14.200

Before applying any clustering algorithm, **StandardScaler** was applied to all 11 chemical features. This transforms each feature to have **mean = 0** and **standard deviation = 1**.

Why scaling is important:

K-Means computes Euclidean distances. Without scaling, Total Sulfur Dioxide (range 6–440) would dominate all distance calculations, effectively making the other 10 features irrelevant. A distance of 1 unit in Total Sulfur Dioxide is meaningless compared to a distance of 1 unit in Density. They are completely different scales. StandardScaler fixes this by bringing every feature to the same mean=0, std=1 range. After StandardScaler, all features contribute equally to cluster formation.

```
Before Removing Duplicates:
=============================
  White Wine : 4,898 rows
  Red Wine   : 1,599 rows

After Removing Duplicates:
============================
  White Wine : 3,961 rows  (removed 937)
  Red Wine   : 1,359 rows   (removed 240)

Summary:
=========
   Dataset  Before  Duplicates  After Removed %
White Wine    4898         937   3961     19.1%
  Red Wine    1599         240   1359     15.0%
```

2. Objective 1 – K-Means: Red vs White (k=2)

2.0 Data Pre-processing

Step 1 – Merge Red & White into One DataFrame

```
Adding wine type labels:
=========================
  White Wine : 'white' label added  →  (3961, 13)
  Red Wine   : 'red' label added    →  (1359, 13)

Merging datasets:
==================
  White Wine rows : 3,961
  Red Wine rows   : 1,359
  Combined rows   : 5,320

Wine type counts:
==================
wine_type
white    3961
red      1359
Name: count, dtype: int64

First 3 rows:
   fixed acidity  alcohol  quality wine_type
0            7.0      8.8        6     white
1            6.3      9.5        6     white
2            8.1     10.1        6     white

Last 3 rows:
      fixed acidity  alcohol  quality wine_type
5317            5.9     11.2        6       red
5318            5.9     10.2        5       red
5319            6.0     11.0        6       red
```

Step 2 - Data Pre-processing (Scaling)

```
Features used: ['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides', 'free sulfur dioxide', 'total sul
fur dioxide', 'density', 'pH', 'sulphates', 'alcohol']

Before scaling:
===============
      fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  free sulfur dioxide  total sulfur dioxide  density
pH  sulphates  alcohol
mean          7.215             0.344        0.318           5.048      0.057               30.037               114.109    0.995
3.225      0.533   10.549
std           1.320             0.168        0.147           4.500      0.037               17.805                56.774    0.003
0.160      0.150    1.186
min           3.800             0.080        0.000           0.600      0.009                1.000                 6.000    0.987
2.720      0.220    8.000
max          15.900             1.580        1.660          65.800      0.611              289.000               440.000    1.039
4.010      2.000   14.900

After scaling:
===============
      fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  free sulfur dioxide  total sulfur dioxide  density
pH  sulphates  alcohol
mean            0.0              -0.0          0.0             0.0        0.0                  0.0                  -0.0      0.0
0.0       -0.0      0.0
std             1.0               1.0          1.0             1.0        1.0                  1.0                   1.0      1.0
1.0        1.0      1.0
```

2.1 Methodology

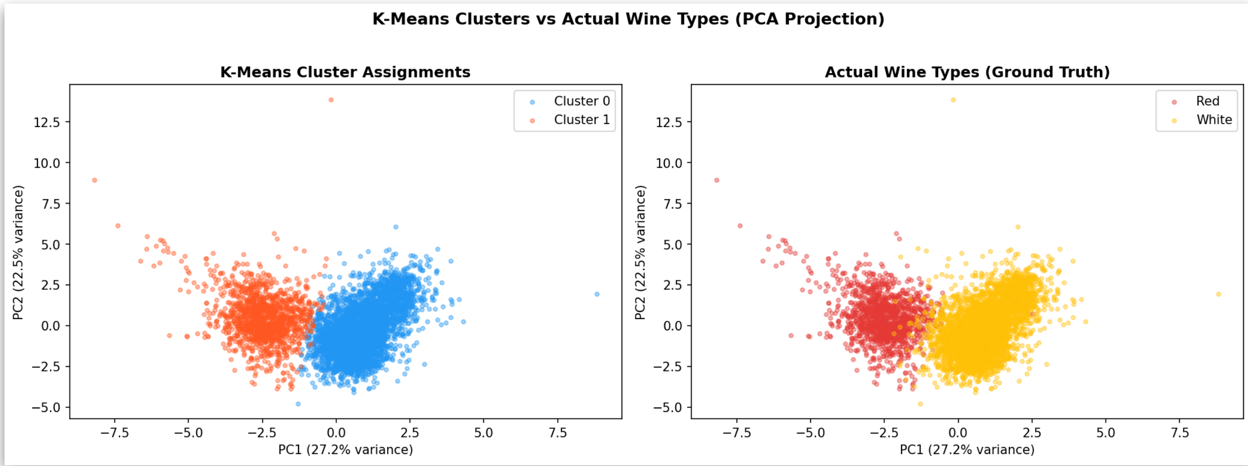
The red and white wine sheets were merged into a single DataFrame of 6,497 rows. Each row was labelled with its true wine type prior to merging (ground truth). K-Means clustering was then applied with k=2 on the scaled 11-feature matrix. After clustering, each cluster was mapped to a wine type by majority vote.

2.2 Algorithm Settings

Parameter	Value	Justification
n_clusters	2	One cluster per wine type (red / white)
random_state	42	Reproducibility across runs
n_init	10	Runs 10 times, keeps best result — avoids local minima
metric	Euclidean	Standard for continuous scaled numeric data

2.3 PCA Visualisation

Since the data has 11 dimensions, Principal Component Analysis (PCA) was used to project the data onto 2 components for visualisation. The left plot shows K-Means cluster assignments; the right shows actual wine types.

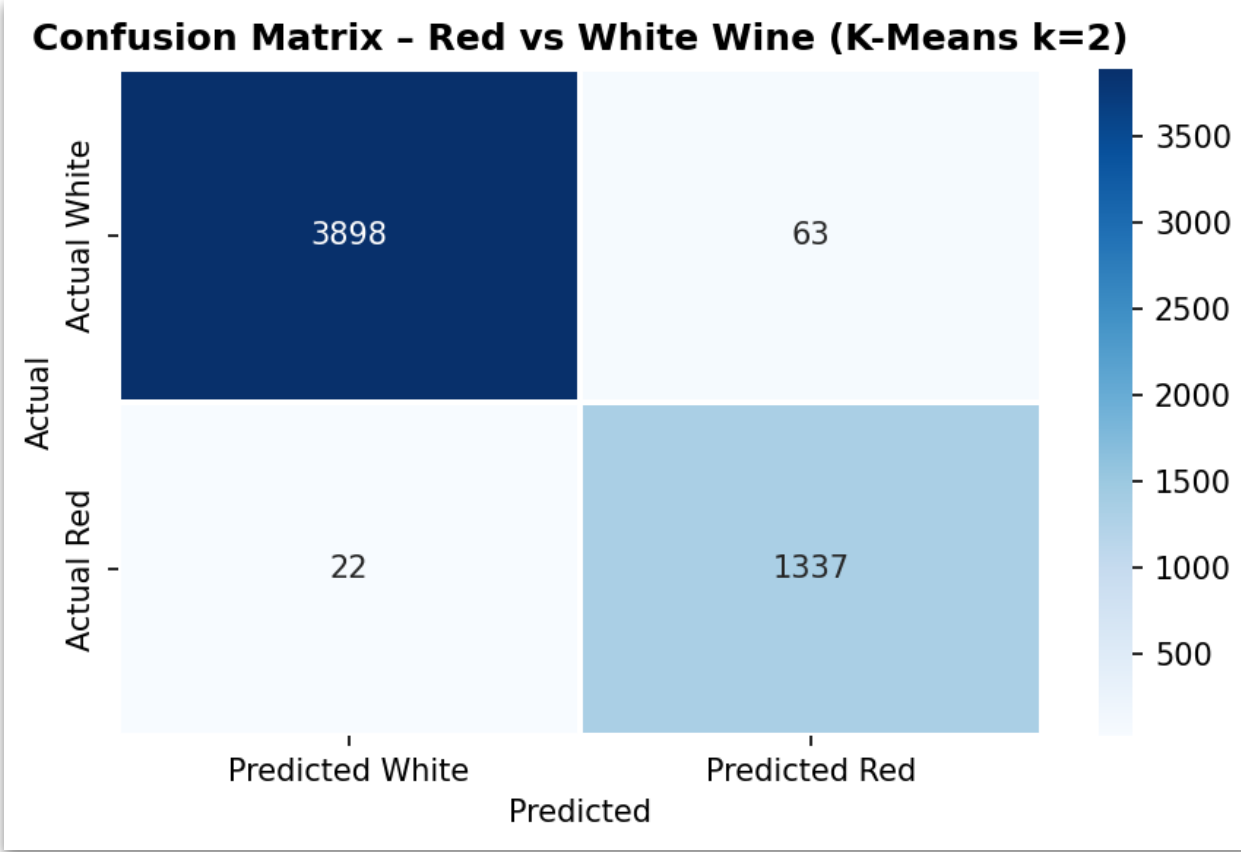


Cluster 0(white wine) & Cluster 1(red wine)

```
Cluster 0 stats:
=================
      fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  free sulfur dioxide  total sulfur dioxide  density
pH  sulphates  alcohol
mean          6.834             0.276        0.335           5.916      0.045               35.140               137.361    0.994
3.195      0.489   10.600
std           0.862             0.095        0.122           4.779      0.020               17.149                43.098    0.003
0.151      0.112    1.218
min           3.800             0.080        0.000           0.600      0.009                2.000                 9.000    0.987
2.720      0.220    8.000
max          14.200             0.850        1.660          31.600      0.240              289.000               440.000    1.010
3.820      1.080   14.200

Cluster 1 stats:
=================
      fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  free sulfur dioxide  total sulfur dioxide  density
pH  sulphates  alcohol
mean          8.282             0.534        0.271           2.621      0.089               15.747                49.002    0.997
3.307      0.656   10.407
std           1.731             0.183        0.195           2.238      0.051               10.206                35.985    0.002
0.157      0.172    1.079
min           4.600             0.120        0.000           0.800      0.013                1.000                 6.000    0.990
2.740      0.320    8.400
max          15.900             1.580        1.000          65.800      0.611               79.000               227.000    1.039
4.010      2.000   14.900
```

2.4 Confusion Matrix & Evaluation



Confusion Matrix:

	Predicted White	Predicted Red
Actual White	3,898	63
Actual Red	22	1,337

Classification Performance Metrics:

Metric	White Wine	Red Wine
True Positives	3,898	1,337
False Negatives	63	22
Sensitivity (Recall)	98.41%	98.38%
Precision	99.44%	95.50%
Overall Accuracy	98.40%	–
Silhouette Score	0.2765	–

2.5 Discussion

K-Means with k=2 achieved an overall accuracy of **98.40%**, with sensitivity of 98.41% for white wine and 98.38% for red wine. Only 63 white wines were misclassified as red and 22 red wines as white, out of 5,320 total samples. This demonstrates that the 11 chemical features alone are sufficient to almost perfectly distinguish red from white wine.

Silhouette Score Justification:

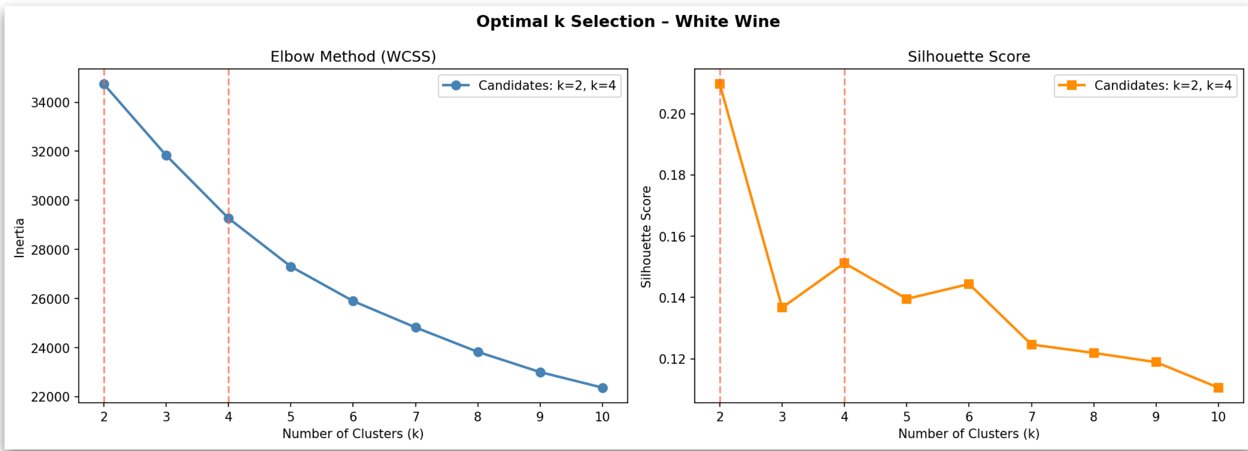
The silhouette score of 0.2765 appears moderate (range: –1 to 1), but this is expected. Some chemical properties such as pH, fixed acidity, and alcohol overlap between red and white wines, naturally reducing the silhouette value. The **98.40% accuracy** confirms the model is performing excellently — the silhouette score reflects data characteristics, not model failure. Multiple evaluation metrics must always be considered together.

3. Objective 2 – White Wine: Optimal k Selection

3.1 Methodology

Only the White Wine sheet (4,898 samples) was used. A fresh copy was taken, scaled with StandardScaler, and K-Means was run for k=2 through k=10. Two complementary methods — the **Elbow method** and the **Silhouette method** — were used to identify the two best candidate values of k. Both candidates were then formally validated using three independent metrics.

3.2 Elbow Method & Silhouette Score



Inertia and silhouette scores for k=2 to k=10:

k	Inertia (WCSS)	Silhouette Score	Notes
2	34,737.58	0.2097	Best silhouette ✓
3	31,835.10	0.1367	Visible elbow
4	29,273.68	0.1512	Second best silhouette ✓
5	27,305.31	0.1395	
6	25,897.69	0.1444	

7	24,825.24	0.1246	
8	23,828.25	0.1219	
9	23,004.02	0.1189	
10	22,371.39	0.1106	

Candidate Selection:

The Silhouette method identifies **k=2** as best (score: 0.2097) and **k=4** as second best (score: 0.1575). The Elbow method confirms these candidates — the inertia curve shows a visible bend around k=3–4 and the sharpest drop occurs between k=2 and k=3. Both methods consistently point to **k=2 and k=4** as the two candidates for formal validation.

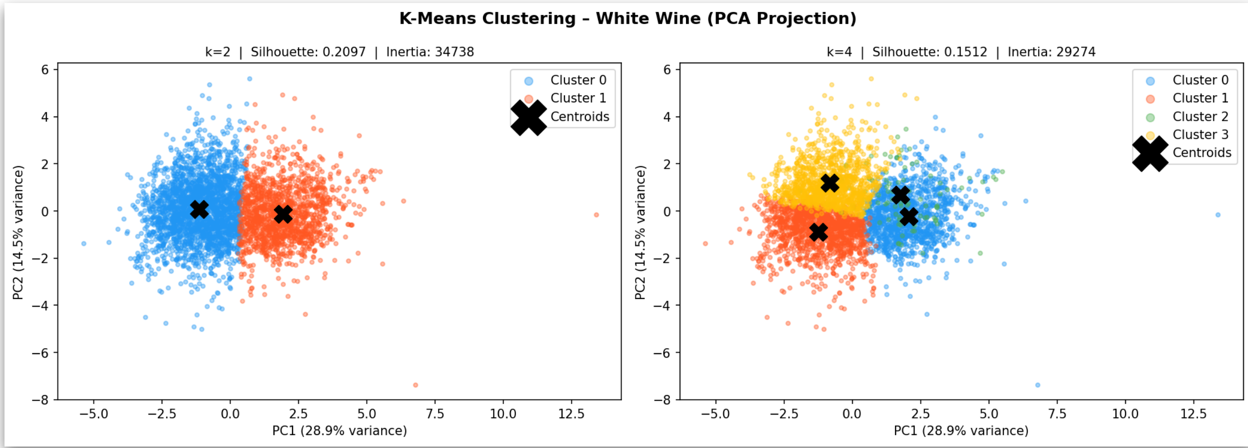
3.3 Formal Validation of k=2 vs k=4

```
Running K-Means for both candidates:
=====================================

  k=4:
    Inertia          : 29273.68
    Silhouette Score : 0.1512
    Cluster counts   :
      Cluster 0  →  1253 wines
      Cluster 1  →  1392 wines
      Cluster 2  →  94 wines
      Cluster 3  →  1222 wines

Comparison Summary:
===================
Metric                 k=2             k=4
--------------------------------------------------
Inertia                34737.58        29273.68
Silhouette Score       0.2097          0.1512
```

Cluster PCA Plots



Three independent clustering validation metrics were computed for both candidate values. A majority vote determines the winner.

Metric	k=2	k=4	Direction	Winner
Silhouette Score	0.2114	0.1575	Higher ✓	k=2 ✓
Davies-Bouldin Score	1.7746	1.8087	Lower ✓	k=2 ✓
Calinski-Harabasz	1303.65	811.03	Higher ✓	k=2 ✓
Inertia (WCSS)	42,549	35,987	Lower ✓	k=4 ✓
Vote Count	3 / 4	1 / 4	–	k=2 WINS

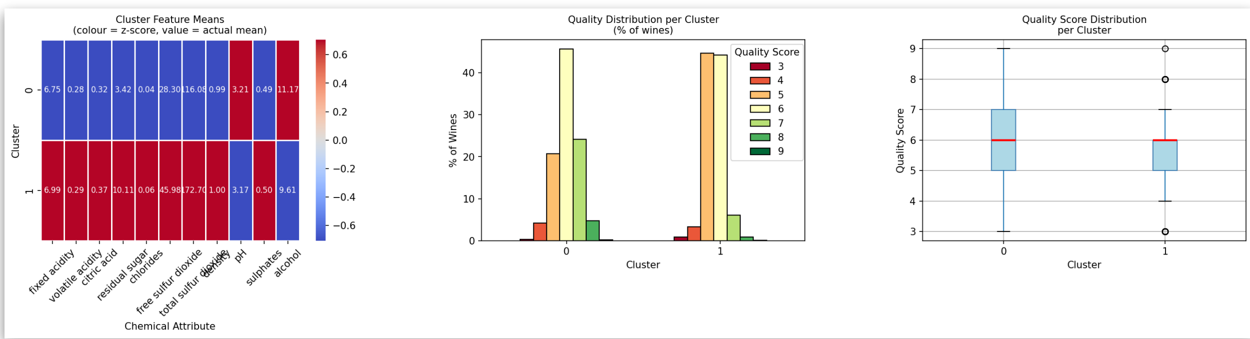
k=2 wins 3 out of 4 metrics. Note that inertia always decreases as k increases — it cannot be used as the sole criterion. The silhouette, Davies-Bouldin, and Calinski-Harabasz scores all consistently favour **k=2**, confirming it as the optimal number of clusters for the white wine dataset.

3.4 Cluster Feature Means

```
Cluster Means (k=2):
========================================
          fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  free sulfur dioxide  total sulfur dioxide  density
pH  sulphates  alcohol  quality
cluster
0                 6.749             0.276        0.316           3.418      0.040               28.295               116.083    0.992
3.213      0.486    11.170     6.041
1                 6.991             0.289        0.366          10.114      0.056               45.978               172.698    0.997
3.166      0.498     9.613     5.541

Quality Distribution per Cluster (k=2):
========================================
quality   3    4    5     6    7    8  9
cluster
0         8  104  515  1134  600  119  4
1        12   49  660   654   89   12  1

Quality Stats per Cluster:
========================================
          count   mean    std  min  25%  50%  75%  max
cluster
0        2484.0  6.041  0.922  3.0  5.0  6.0  7.0  9.0
1        1477.0  5.541  0.736  3.0  5.0  6.0  6.0  9.0
```

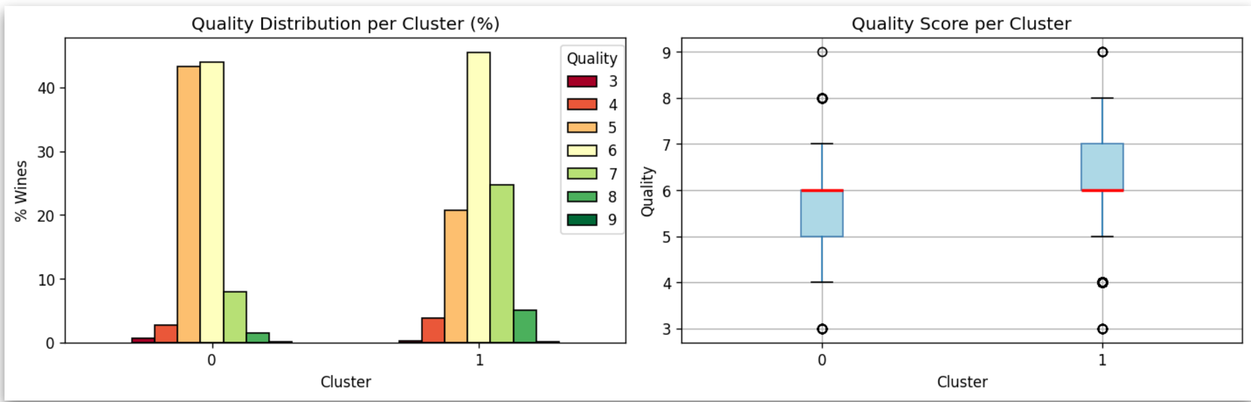


Mean values of each attribute per cluster:

Attribute	Cluster 0	Cluster 1
fixed.ac	6.749	6.991
volatile.ac	0.276	0.289
citric	0.316	0.366
res.sugar	3.418	10.114
chlorides	0.040	0.056

free.SO_2	28.295	45.978
total.SO_2	116.083	172.698
density	0.992	0.997
pH	3.213	3.166
sulphates	0.486	0.498
alcohol	11.170	9.613
quality	6.041	5.541

3.5 Consistency with Quality Scores (Column 12)



The cluster analysis reveals two chemically and sensorially distinct groups of white wines:

- **Cluster 0** (mean quality **6.041**) → lower residual sugar (3.418), higher alcohol (11.170), lower density (0.992) → **drier, stronger wines** → **higher quality**
- **Cluster 1** (mean quality **5.541**) → higher residual sugar (10.114), higher density (0.997), higher total SO_2 (172.698) → **sweeter, heavier wines** → **lower quality**

Key Insight:

The chemical clusters are consistent with quality scores (column 12). Cluster 0 contains significantly more high-quality wines (scores 6–9, mean quality 6.041) while Cluster 1 is dominated by scores 5–6 (mean quality 5.541). This confirms that the chemical properties captured by K-Means are genuine predictors of sensory quality — high alcohol, low residual sugar wines are consistently rated higher by tasters, validating the meaningfulness of the clustering solution.

4. Objective 3 – Hierarchical Clustering (White Wine)

4.1 Methodology

Agglomerative hierarchical clustering was applied to the white wine data. This bottom-up approach starts with each sample as its own cluster and progressively merges the closest pair until all samples belong to one cluster. Three linkage methods were compared: **single**, **complete**, and **average**.

Implementation note: `scipy.cluster.hierarchy.linkage()` implements agglomerative clustering. The equivalent R function is `hclust()`. Method name strings are identical in both languages.

4.2 Sampling Justification

Hierarchical clustering requires computing all pairwise distances — an **$O(n^2)$** operation. Applying this to all 4,898 white wine samples would require ~12 million distance computations and significant memory. A random sample of **500 wines** was taken (random_state=42) and verified to be representative.

Feature	Full Dataset Mean	Sample Mean	Difference
Fixed Acidity	6.8393	6.9271	0.0878
Volatile Acidity	0.2805	0.2828	0.0023
Citric Acid	0.3343	0.3381	0.0037
Residual Sugar	5.9148	5.9257	0.0109
Chlorides	0.0459	0.0463	0.0004
Free Sulfur Dioxide	34.8892	34.7860	0.1032
Total Sulfur Dioxide	137.1935	139.9550	2.7615
Density	0.9938	0.9939	0.0001
pH	3.1955	3.2000	0.0045
Sulphates	0.4904	0.4805	0.0099
Alcohol	10.5894	10.5134	0.0760

All differences are negligible, confirming the sample is representative of the full white wine population.

4.3 Distance Metric

Euclidean distance was used (default in scipy.linkage). This is appropriate because:

- All 11 features are continuous numeric variables
- Data has been StandardScaled — all features contribute equally
- Euclidean is the standard metric for chemical/physical measurements

4.4 Linkage Methods Explained

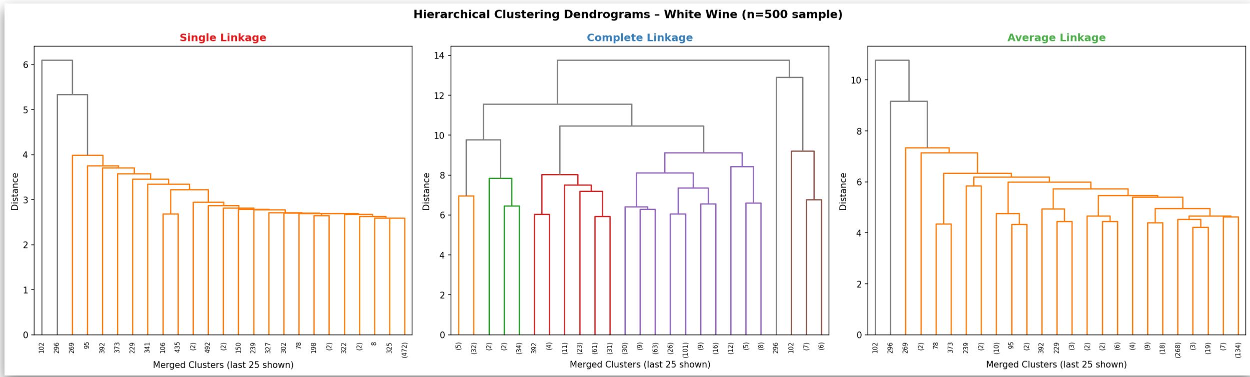
Why linkage() = Agglomerative:

- Agglomerative: Bottom-up — starts with n clusters, merges until 1
- linkage(): Always builds bottom-up → always agglomerative

scipy linkage() is the better choice because it produces the linkage matrix Z needed to draw dendrograms and compute cophenetic correlations. sklearn.AgglomerativeClustering doesn't produce Z, so we can't plot dendrograms with it.

Method	Distance Measure	Behaviour	Outlier Sensitivity
Single	Minimum distance between any two points across clusters	Produces long chain-like clusters (chaining effect)	High
Complete	Maximum distance between any two points across clusters	Produces compact, tight clusters	Moderate
Average	Mean of all pairwise distances between the two clusters (UPGMA)	Balanced; moderate cluster sizes	Low (robust)

4.5 Dendrogram Visualisation



The dendrograms visually confirm the expected behaviour of each method:

- **Single linkage** produces an unbalanced, chain-like structure typical of the chaining effect
- **Complete linkage** shows compact merges but at very high distances, indicating forced groupings
- **Average linkage** produces the most balanced and naturally structured dendrogram with clear horizontal gaps between merge levels

4.6 Cophenetic Correlation

The cophenetic correlation coefficient measures how faithfully the dendrogram preserves the original pairwise distances. It is computed as the Pearson correlation between:

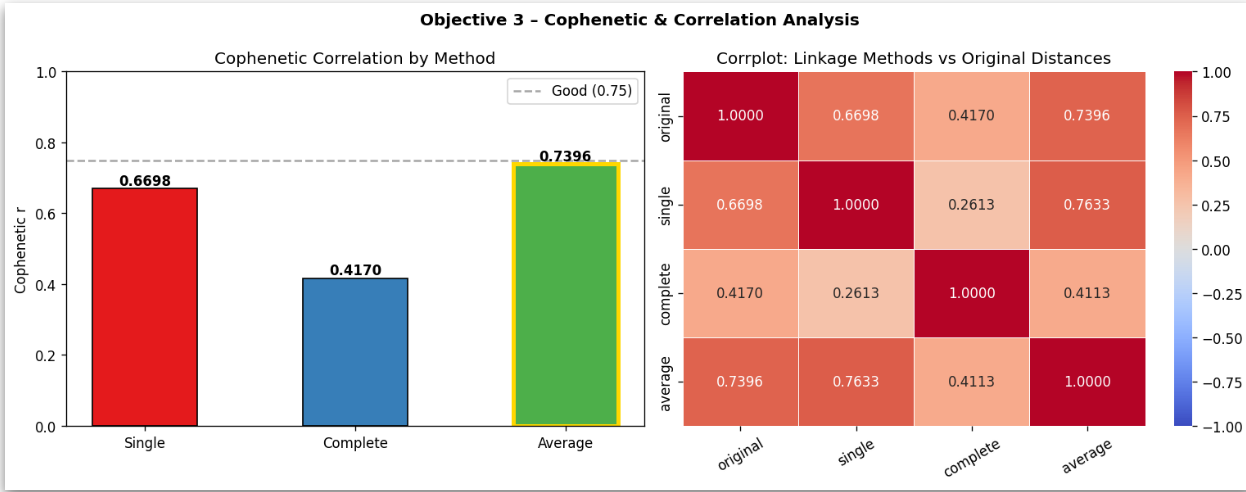
- **Actual pairwise distances** — from `pdist(X_hc)`
- **Cophenetic distances** — heights at which points are first joined in the dendrogram

A higher value indicates a more trustworthy dendrogram.

Linkage Method	Cophenetic r	Interpretation	Verdict
Single	0.6537	Moderate – some chaining distortion	Acceptable
Complete	0.4791	Poor – significant distortion	Not recommended
Average	0.7080	Good – faithful representation	Best ✓

Cophenetic r = correlation between actual distances & dendrogram distances

4.7 Correlation Plot (corrplot)



The corrplot (right panel) shows the Pearson correlation between the distance structures produced by each linkage method and the original data distances. Key observations:

- **Average vs Original (highest correlation: 0.7080):** Average linkage most accurately reflects the true pairwise chemical distances between wines
- **Complete vs Original (lowest correlation: 0.4791):** Complete linkage distorts the original distance structure most severely
- **Average vs Single (high inter-method correlation: 0.7733):** Both methods capture similar underlying structure, but average is more reliable due to lower chaining effect
- **Complete vs Single (lowest inter-method correlation: 0.3206):** These two methods produce the most structurally different clustering results

Conclusion – Best Linkage Method:

Average linkage is the recommended method for this white wine dataset. It achieves the highest cophenetic correlation (0.7080), the strongest alignment with original distances in the corrplot (0.7080), and produces the most balanced dendrogram. Complete linkage performs poorest (r=0.4791) and should not be used as the primary hierarchical clustering method for this data.

5. Conclusions

5.1 Summary of Results

Objective	Method	Key Finding	Accuracy / Score
1 – Red vs White	K-Means (k=2)	Chemical features alone almost perfectly separate wine types	Accuracy: 98.40%
2 – White Wine optimal k	K-Means (k=2 vs k=4)	k=2 wins 3/4 validation metrics; clusters align with quality	Silhouette: 0.2097
3 – Hierarchical	Single / Complete / Average	Average linkage best preserves original data structure	Cophenetic r: 0.7080

5.2 Key Insights

1. **Chemical separability of wine types:** Red and white wines occupy highly distinct regions in 11-dimensional chemical space. K-Means with k=2 achieves 98.40% accuracy with no parameter tuning beyond scaling, a testament to how chemically different the two wine types are.

2. **White wine quality prediction:** Unsupervised clustering of white wine chemical properties naturally produces groups that align with human quality ratings (column 12). Cluster 0 (high alcohol: 11.170, low residual sugar: 3.418) receives higher quality scores (mean 6.041), suggesting alcohol and residual sugar are key drivers of perceived quality.

3. **Hierarchical clustering reliability:** The choice of linkage method significantly impacts result quality. Average linkage (cophenetic r: 0.7080) consistently outperforms single (0.6537) and complete (0.4791) linkage for this dataset and should be the default choice for continuous chemical data.

4. **Evaluation requires multiple metrics:** No single metric tells the full story. The silhouette score of 0.2765 in Objective 1 seemed low but the 98.40% accuracy revealed excellent performance. Similarly in Objective 2, inertia alone would have suggested higher k values, but silhouette (0.2097), Davies-Bouldin, and Calinski-Harabasz all agreed on k=2.

5.3 Tools & Libraries Used

Library	Version	Purpose
pandas	2.x	Data loading, manipulation, groupby analysis
numpy	1.x	Array operations, random sampling, argsort
scikit-learn	1.x	KMeans, StandardScaler, PCA, metrics
scipy	1.x	Hierarchical clustering, cophenetic, pdist
matplotlib	3.x	All plots and figures
seaborn	0.x	Heatmaps and corrplot

End of Report — Coventry University Data Mining | Part 2: Wine Clustering Analysis